

Report - Design Analysis Process + Physical ERD + Data Dictionary completed

Studio 1 project: NestUp

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Section 1: The Design Analysis Process

To transition the messy real-world operations of a real estate agency into an optimized relational schema, the system design followed a structured three-tier investigative data modeling framework: Exhaust → Elimination → Define.

1.1 The "Exhaust" Phase (Domain Brainstorming): We started by writing down a big brainstorm list of every data point, attribute, physical asset, and business transaction involved in the daily workflow of a property agency: Agent name, agent mobile contact, commission tier percentage, house key tracking tags, client text messages, phone call recordings, vendor name, vendor phone number, house street address, valuation pricing, layout room configurations, total land square meters, lawn maintenance checklists, buyer name, buyer maximum budget, neighborhood preferences, viewing appointment calendar timestamps, buyer feedback commentary, structural engineers' reports, offer bids, down-payment parameters, contract settlement timestamps.

1.2 The "Elimination" Phase (System Scoping): Next we went through our brainstorm list of every data point against strict database system boundaries. Items that were purely physical, highly fluid, or legally untraceable were eliminated: House Keys & Lawn Signage: Eliminated. These represent physical inventory assets handled via direct manual operations, not database data fields. Text Messages & Phone Calls: Eliminated. Dynamic, unstructured daily communications belong inside external telecom networks, not fixed relational rows. Structural Engineers' Reports: Eliminated. We left these big reports out of the database because they can be stored on a normal Google drive instead.

1.3 This step generated our 7 foundational system themes:

Agent: Staff directory and transaction rate tracking.

Vendor: Client portfolios listing properties for sale.

Buyer: Client profiles actively hunting for matching listings.

Property: Structural asset details and current listing state values.

Viewing: Schedule tracker resolving agent/buyer inspections.

Offer: Legal financial negotiation step logs.

Sale: Final contract accounting entries.

Section 2: Physical Entity-relationship Diagram (ERD)

This section translates the NestUp problem domain into an explicit relational database architecture using Crow's Foot Notation logic. It implements Primary Keys (PK) to guarantee identity uniqueness and places Foreign Keys (FK) on the "many" side of relationships to guarantee structural referential integrity.

2.1 System Relational Mapping Rules:

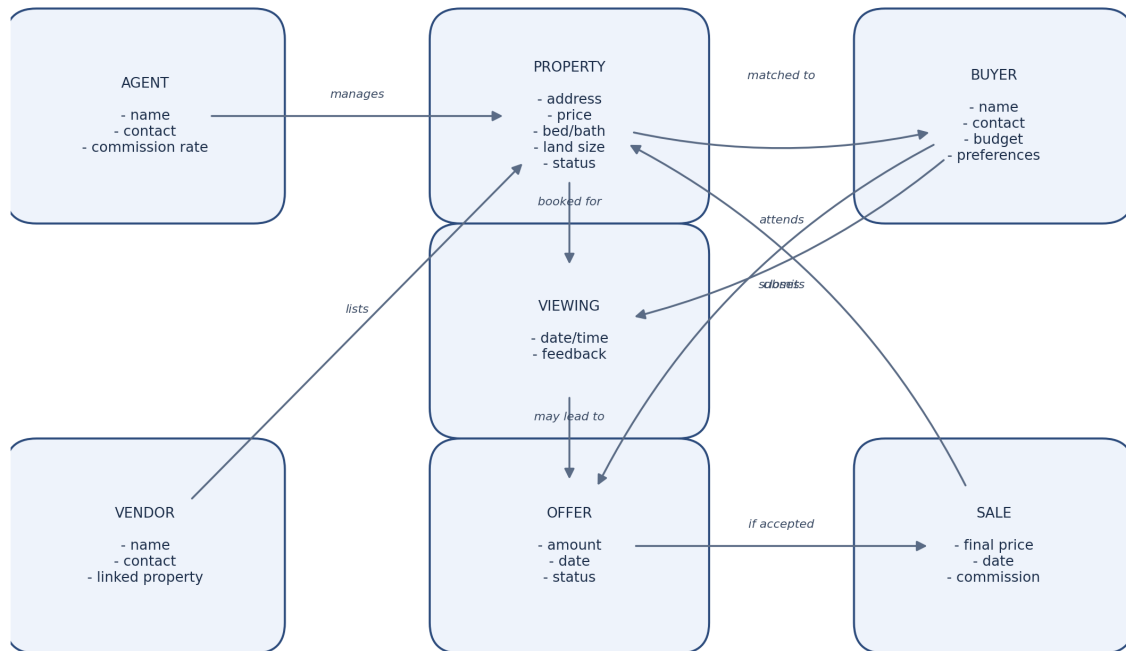
AGENT to PROPERTY: An agent coordinates a portfolio of one or many properties. A property has exactly one handling agent. Relational consistency is locked by inserting agent_id into the PROPERTY table.

VENDOR to PROPERTY: A property must be listed by exactly one vendor. A single vendor may list multiple separate properties. Linked via vendor_id in the PROPERTY table.

BUYER and PROPERTY to VIEWING (Associative Resolution): A buyer visits many properties, and a property hosts many distinct buyers. VIEWING breaks down this many-to-many relationship smoothly by storing property_id and buyer_id as foreign keys.

OFFER to SALE: To ensure a single listing is never double sold, the SALE ledger binds offer_id under a strict UNIQUE flag constraint.

2.2 System Architecture Sketch:



Section 3: Completed Data Dictionary

This section provides the technical schema specifications for the NestUp relational storage engine, defining data types, primary keys (PK), foreign keys (FK), and data validation boundaries.

Table 1: agent (Internal directory tracking sales personnel profiles)

• *Primary Key:* `agent_id`

Column Name	Data Type	Constraints	Description / Business Rule
<code>agent_id</code>	INTEGER	PRIMARY KEY AUTOINCREMENT	Auto-generated unique representative key.
<code>name</code>	TEXT	NOT NULL	Full legal first and last name.
<code>contact</code>	TEXT	NOT NULL	Corporate electronic email address string.
<code>commission_rate</code>	REAL	NOT NULL CHECK(<code>commission_rate</code> <= 10.0)	Base percentage tier earned per transaction.

Table 2: vendor (Profile records for individuals listing assets for sale)

- *Primary Key: vendor_id*

Column Name	Data Type	Constraints	Description / Business Rule
vendor_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System-allocated key for internal operational routing.
name	TEXT	NOT NULL	Full legal name of the seller client.
contact	TEXT	NOT NULL	Primary mobile contact phone line.

Table 3: buyer (System preferences profile logs for looking up active prospects)

- *Primary Key: buyer_id*

Column Name	Data Type	Constraints	Description / Business Rule
buyer_id	INTEGER	PRIMARY KEY AUTOINCREMENT	System internal tracking token key.
name	TEXT	NOT NULL	Full name of client seeking real estate.
contact	TEXT	NOT NULL	Mobile telephone line or electronic text address.
budget	INTEGER	NOT NULL CHECK(budget > 0)	Maximum buying financial layout limit (NZD).
preferences	TEXT	NULL	Area neighborhood target matching text keywords.

Table 4: property (Master record tracking active structural listings)

- *Primary Key: property_id*

Column Name	Data Type	Constraints	Description / Business Rule
property_id	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique listing identification number.
address	TEXT	NOT NULL UNIQUE	Geographically verified street entry.

price	INTEGER	NOT NULL CHECK(price > 0)	Public marketing listed price tag (NZD).
bed_bath	TEXT	NOT NULL	Metric layout descriptor code (e.g. '4B/3B').
land_size	INTEGER	NOT NULL	Measured square meters size footprint.
status	TEXT	DEFAULT 'Listed'	Pipeline tracking: 'Listed', 'Under Offer', 'Sold'.
vendor_id	INTEGER	FOREIGN KEY REFERENCES vendor	Bridges back to parent owner in vendor table.
agent_id	INTEGER	FOREIGN KEY REFERENCES agent	Bridges back to parent representative in agent table.

Table 5: viewing (Mapping table linking buyers to listed site inspections)

• *Primary Key: viewing_id*

Column Name	Data Type	Constraints	Description / Business Rule
viewing_id	INTEGER	PRIMARY KEY AUTOINCREMENT	Sequential schedule log row entry token.
property_id	INTEGER	FOREIGN KEY REFERENCES property	Reference tracking back to target property listing.
buyer_id	INTEGER	FOREIGN KEY REFERENCES buyer	Reference tracking back to attending buyer.
date_time	TEXT	NOT NULL	Inspection timestamp using standard ISO 8601 string.
feedback	TEXT	NULL	Free-form review comments uploaded by staff.

Table 6: offer (Ledger tracking monetary values presented by buyers)

• *Primary Key: offer_id*

Column Name	Data Type	Constraints	Description / Business Rule
offer_id	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique formal negotiation item index.
property_id	INTEGER	FOREIGN KEY REFERENCES property	Link pointing to targeted property item.
buyer_id	INTEGER	FOREIGN KEY REFERENCES buyer	Link pointing to offering client party.
amount	INTEGER	NOT NULL CHECK(amount > 0)	Explicit financial offer bid in NZD.
date	TEXT	NOT NULL	Logged presentation date string notation.
status	TEXT	DEFAULT 'Pending'	Evaluation state: 'Pending', 'Accepted', 'Declined'.

Table 7: sale (Accountant balancing ledger storing finalized closures)

• *Primary Key: sale_id*

Column Name	Data Type	Constraints	Description / Business Rule
sale_id	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique final balancing reference token.
offer_id	INTEGER	NOT NULL UNIQUE FOREIGN KEY REFERENCES offer	Strictly locks down 1:1 accepted offer validation.
final_price	INTEGER	NOT NULL CHECK(final_price > 0)	Definitive transactional settled payout value.
date	TEXT	NOT NULL	Deed settlement title sign-off timestamp.
commission	REAL	NOT NULL	Payout value matching final_price * agent rate.